



OPEN An adaptive emotion aware music education system using machine learning for personalized emotional wellbeing enhancement

Xinyun Deng¹, Yongxin Zhou^{2✉}, Xingzhi Guan³ & Yuhui Wang¹

Music education is crucial to cognitive and emotional well-being. Nevertheless, the traditional approaches are not personalized and are unable to dynamically respond to the affective states of the learners, restricting their influence on the emotional interest. As a solution to this void, we will design an Adaptive Emotion-Aware Music Education System (AEMES) which can use the techniques of machine learning to monitor, model and respond to the real-time emotions of learners. The originality of this framework is that it is based on an integrated combination of emotion modeling, adaptive strategy selection, and personalized intervention mechanisms that allow the system to achieve continuous engagement of learners and promote emotional wellbeing. The suggested approach is strictly tested on a variety of datasets that include multimodal responses of learners, which include physiological data, facial expressions and behavioral interactions. In comparison with state-of-the-art methods that include CNN-LSTM, Audio-Transformer, and Visual-Transformer models, it is shown that AEMES performs much better than the existing methods in standard performance measures. The proposed framework has a quantitative performance of an accuracy of 87.2%, F1-score of 85.7% and AUROC of 0.91 compared to baseline models, which have a margin of over 4–8% across measures. The role of emotion modeling and adaptive strategies to the overall performance of the system is also confirmed by ablation studies. Emotional pathways (both in terms of time trends and multi-emotional interactions) visualization illustrates the possibility of the framework to dynamically control emotion of learners and ensure that it does not go negative during learning sessions. The overall results support the claim that AEMES is effective in providing personalized, emotion-sensitive music education, which would prove to be a powerful method of improving the learning process and emotional wellbeing.

Keywords Emotion-aware music education, Adaptive learning systems, Affective computing, Machine learning, Emotional wellbeing

Music has always been perceived as an effective tool of expressing emotions, cognitive growth, and individual health. The learning of music in educational institutions is not only linked to the attainment of artistic and technical skills, but also the attainment of emotional awareness, creativity as well as psychological stability. Numerous teachers and scholars admit that music may affect mood, decrease stress and improve emotional control in students. Nevertheless, the old structure of teaching music provides a fixed way of teaching that cannot be used to consider individual emotional condition, learning styles and psychological requirements. Consequently, students who have dissimilar emotional reactions to music might not enjoy the learning process in totality. The recent years have seen the fast development of machine learning and artificial intelligence, which has created a new field of creating adaptable and smart learning spaces. These systems allow complex user behavior, physiological signal and multi-media data analysis so that each person can receive personalized experiences. Machine learning can be applied to music education where the emotional reactions to music are analyzed and educational content is adjusted. With the incorporation of emotion recognition and music learning systems, one can develop systems to dynamically react to the emotional state of a learner and develop learning materials that can deliver not only the skill but also the emotional state¹.

¹School of Architecture and Art, Central South University, Changsha 410083, Hunan, China. ²School of Music, Theatre & Dance, University of Michigan, Ann Arbor, MI 48109, USA. ³Faculty of Social Sciences & Liberal Arts, UCSI University, Kuala Lumpur 56000, Malaysia. ✉email: yongxinzhou091@gmail.com

Emotion-conscious learning systems will attempt to reconcile the disjuncture existing between cognition in learning and affect. Emotions are important in the results of learning since they determine attention, motivation, retention of memory and interest. As an illustration, positive emotional states are usually more conducive to creativity and focus whereas negative emotional states like stress or anxiety tend to prevent effective learning. Within music education, in which expressing emotions is an essential aspect of artistic work, the ability to identify and address the emotional state of students can greatly contribute to the success of the teaching process. Although the significance of affective factors has been realized, not all of the current digital music learning systems are designed to deliver content and assess performance as their core functions, and have little ability to track or react to the emotional experience of the learners. To overcome this limitation, scholars have started to consider how affective computing methods can be used in music education systems. Affective computing is concerned with allowing machines to identify, understand, and react to human emotions with a range of data types of facial expressions, physiological cues, and audio patterns. These signals can be analyzed using machine learning algorithms to determine such emotional states as happiness, sadness, relaxation, or stress. As part of the music education systems, such capabilities enable systems to suggest music exercises, learning contents, or therapeutic listening sessions that are compatible with the emotional state of the learner².

This work was inspired by these advancements and suggests an adaptive emotion-aware music learning system that uses machine learning to customize the learning process and improve wellbeing by promoting emotional experience. The suggested structure would be used to track the emotional reactions of the learners to music and actively modify the instructional materials to help them achieve musical proficiency and psychological balance. The system intends to provide a more interactive and receptive environment of learning music by combining both emotional recognition systems and smart recommendation systems. Finally, it is also aimed at explaining how emotion-sensitive technologies will help to transform traditional music education into a more personalized learning process that will not only result in musical skills but also in emotional fitness.

Problem statement

Although the positive aspects of music education have already been documented extensively, the existing digital learning systems do not incorporate real-time emotion recognition and adaptive intervention measures. It is not unusual to find that learners will have an unstable level of emotions, between engagement and excitement and frustration and disengagement during music learning sessions. Traditional methods do not have the systems that would track, simulate, and react to these affective states that are dynamic. As a result, students might feel demotivated, have low retention, and emotional dissatisfaction. This issue is increased in the heterogeneous classes where the emotion reactions of different age groups, cultural groups, and the previous musical knowledge differ greatly³.

The main issue of the current study is that of designing an individualized, emotion-sensitive music education system that can constantly monitor the emotional cues, simulate the affective state, and provide the adaptive intervention that facilitates the engagement and the development of emotional wellbeing. The system must not only enhance learning outcomes but also positive emotional experiences in the long-term⁴.

Motivation

Recent developments in affective computing and multimodal machine learning offer successful solutions to both record and read human emotions based on physiological indicators, facial expressions, voice patterns, and behavioral displays. When applied to music education, these abilities provide the ability to develop not only the responsive systems but also those which can forecast emotional patterns. With the help of these technologies, one can take the engagement of learners, streamline the teaching materials, and offer emotional encouragement throughout the difficult stages of learning⁵.

This study is driven by the fact that emotional wellbeing is an aspect of successful learning. Music education stands out among other cognitive domains in that it is an area of study that arouses sentimental feelings. Systems that overlook the affective feedback run the risk of losing the interest of learners or not fulfilling the emotional needs of the learners. With the presentation of adaptive and emotion-sensitive systems, one can develop a system of whole-person learning that can foster the development of skills and emotional fulfilment at the same time⁶.

Objectives of the research

This research aims to address existing gaps in digital music education by developing an Adaptive Emotion-Aware Music Education System (AEMES). The primary objectives are:

- **Emotion Modeling:** Accurately capture and model learners' emotional states using multimodal data sources, including physiological, behavioral, and facial cues.
- **Adaptive Intervention:** Dynamically select and deliver personalized learning strategies based on real-time emotional feedback.
- **Performance Validation:** Quantitatively demonstrate that the proposed system outperforms existing state-of-the-art methods across multiple metrics and datasets.
- **Visualization of Emotional Dynamics:** Provide intuitive visualizations of emotion trajectories to validate the effectiveness of the adaptive interventions.

Through these objectives, AEMES seeks to establish a novel, integrated framework for emotion-aware adaptive music education, combining predictive emotion modeling with real-time learning personalization.

Research questions

The study is guided by the following key research questions:

- How can real-time emotional states be accurately captured and modeled in a music learning environment⁷?
- To what extent can adaptive interventions based on emotional feedback enhance learner engagement and emotional wellbeing⁸?
- How does the proposed system compare with state-of-the-art models in terms of classification performance, generalizability, and robustness across diverse datasets⁹?
- What visualization techniques can effectively communicate temporal emotional dynamics and validate system interventions¹⁰?

Significance of the research

Its transdisciplinary nature makes this research valuable. The research uses music pedagogy, emotional computing, and machine learning to create a technologically sophisticated and pedagogically sound framework. The results show:

- **Enhanced Learner Engagement:** By responding to emotional cues, the system maintains sustained motivation and attention.
- **Improved Emotional Wellbeing:** Adaptive interventions reduce frustration and support positive emotional experiences.
- **Quantitative Performance Improvements:** The system demonstrates superior results in classification metrics such as Accuracy, F1-Score, and AUROC, providing empirical validation.
- **Practical Applications:** The framework can be deployed in schools, online learning platforms, and personalized music tutoring environments.

This research contributes to the broader research area of affective learning technologies, providing a blueprint for integrating real-time emotion awareness into educational systems beyond music.

Novelty of the proposed approach

The suggested AEMES framework proposes several novelties:

- **Integrated Multimodal Emotion Modeling:** Unlike the previous studies, which use only one of the modalities, AEMES takes into consideration physiological, behavioral, and facial information in order to provide powerful emotion recognition.
- **Adaptive Strategy Selection:** The system automatically chooses interventions, depending on real-time emotional feedback, to allow individualized and situational learning.
- **Emotional Dynamics:** The visualization of emotional dynamics is done in the form of a temporal affect emotion progression and multi-emotion dynamics to offer interpretable data on affect in the learner, and this has not been addressed in previous literature.
- **Comprehensive Validation:** Extensive experiments across multiple datasets demonstrate the generalizability and superiority of AEMES compared to existing CNN-LSTM, Audio-Transformer, and Visual-Transformer models.

These aspects collectively distinguish AEMES from prior music education and affective computing frameworks, establishing it as a novel contribution to adaptive learning technologies.

Structure of the paper

The rest of the paper is structured in the following way: Sect. 2 is a review of related work in emotion-aware education and adaptive music systems. Section 3 provides the suggested AEMES framework, which consists of emotion modeling, adaptive strategy formulation, and system architecture. In Sect. 4, the datasets, preprocessing, and experiment set are described. In addition, talks about findings and interpretation, such as ablation experiments, comparative analysis, and depiction of emotional time courses. Section 5 provides a conclusion of the study and outlines the future research directions.

Related works

Music education and interactive learning-based emotion-aware systems in music education, particularly in interactive learning, have received significant interest over the last few years, in large part because of progress in affective computing and machine learning. The educational process of music is emotional in nature, and the level of involvement of a learner is strongly connected with such affective states as interest, frustration, excitement, and calm. The online digital music platforms do not pay much attention to the real-time emotional status of the learners, but their main aim is the delivery of the information and skill evaluation. Therefore, students will feel demotivated and experience low levels of emotional satisfaction, which will adversely affect their academic performance. Affective intelligence has come out as one of the solutions to this problem, and integrating it into the educational systems is promising¹¹.

Unimodal music emotion recognition

Music Emotion Recognition (MER), in its initial investigations, concentrated more on the audio characteristics. Scholars derived spectral, temporal, and rhythmic features of music samples to categorize emotions with established machine learning techniques, Support Vector Machines (SVM), k-Nearest Neighbors (kNN), and decision trees^{12,13}. These models were partially successful in classifying fundamental emotions or pairs of valence-arousal and had shortcomings including failure to handle learner-specific feedback or multimodal responses to emotion. The subjectivity of emotional perception and variability of emotional reactions among

different people proves to be a challenge to the performance of unimodal systems, which prompts a shift to more individual and multimodal methods. Deep learning has transformed MER research using CNNs and RNNs¹⁴. CNNs are good in capturing local spectral and temporal information from audio recordings, whereas RNNs and LSTM¹⁵ networks may discern evolving emotional patterns. Despite these developments, most deep learning MER research used music elements without learner response like facial expression or physiological input. These approaches are useful for broad emotion classification but not for educational environments that need fast customization and emotional tweaking¹⁶.

Multimodal emotion recognition

Since unimodal MER is inadequate, current research focuses on multimodal emotion recognition using auditory, visual, and physiological inputs. CNN facial expression analysis, eye-tracking, and posture recognition have been combined with audio characteristics to improve emotion inferences. Heart rate, galvanic skin reaction, and EEG contribute to recognition fidelity, which includes tiny emotional fluctuations that cannot be seen¹⁷. These are more accurate and dependable, sometimes as high as 88–90% for music classification tasks using multimodal fusion methodologies such as early feature concatenation or late decision-level fusion¹⁸. CNN-based audio feature extractors and visual emotion classifiers are used in interactive music teaching systems. This method proved that the combination of various modalities offers a deeper insight into the effect of learners, which allows making the content more adaptable. Likewise, multimodal fusion based on attention mechanisms enables the system to dynamically weight per modality importance to improve real-time emotion inference and prevent sensitivity to noisy data or missing data¹⁹.

Emotion-aware music recommendation systems

Similar to MER, emotion-aware music recommendation systems have also been created to customize music recommendations according to the estimated emotional states. These systems usually apply deep learning to map user emotions to valence-arousal representations of music, and combine collaborative and content-based filtering. The models enhance user experience and satisfaction by proposing music that matches or adapts to the existing emotive conditions. Although effective in recommendation tasks, these methods usually do not have a continuous adaptive intervention capability as required in the educational task. In addition, they fail to consider real-time feedback on learning, which plays a vital role in improving both cognitive and emotional effects in the process of structured music lessons²⁰.

Emotion-aware educational systems

Affective computing has also been utilized in general educational technologies, in addition to music-specific fields. Visual, auditory, or physiological inputs are used to assess student interest, dissatisfaction, or boredom and change teaching or learning. Research suggests that emotion-sensitive learning environments increase learning and well-being. Based on student dissatisfaction, adaptive tutoring systems change problem complexity, enhancing retention and engagement. While mathematics and language learning are universal notions, such frameworks neglect music-specific emotional dynamics, which are more complex and subjective²¹.

Identified research gaps

Critical literature evaluation shows various gaps:

- **Lack of Real-Time Adaptive Interventions:** Most MER and recommendation systems focus on offline emotion prediction without continuous adaptation to dynamic learner states.
- **Limited Multimodal Integration in Education:** While multimodal emotion detection enhances accuracy, few systems use it for tailored music learning modifications.
- **Sparse Visualization and Interpretability:** Existing systems rarely provide interpretable visualizations of emotion trajectories, making it difficult to understand the impact of adaptive interventions.
- **Multiple Dataset Evaluation:** Studies validated on single datasets or restricted modalities may not be generalizable across learners, cultures, and learning situations.

A comprehensive, adaptive framework with multimodal emotion modeling, real-time intervention tactics, interpretable visualization, and rigorous assessment across datasets is needed to close these gaps. Table 1 summarizes the limitations of several existing works.

The studies³² done recently emphasize the increase in the importance of emotional aspects including motivation, self-efficacy, and emotional control in learning among students especially in physically challenging subjects like Physical Education. An investigation on 409 undergraduate PE students in China revealed that AI-enabled affective computing substantially enhanced these psychological outcomes with the most obvious effect on the self-efficacy. The findings indicate that student-centred and adaptive learning environments that improve student engagement and wellbeing can be facilitated by AI systems that are emotionally responsive. In³³, author introduced an Emotion-Aware Intelligent Tutoring System (EA-ITS) that supports learning through emotional awareness of students by recognizing the emotion on the face of students via Convolutional Neural Networks (CNNs) based facial emotion recognition (FER). It varies the content difficulty, feedback and pace according to emotions detected in the system. Findings indicate that emotion recognition is high and learners become motivated, satisfied, and perform better than when using conventional ITS.

The literature has shown that emotion recognition in music and the education field has changed to deep and multimodal architectures as opposed to unimodal and shallow models, improving the classification performance and robustness. The approaches based on combining audio, visual, and physiological signals are more representative of the affective states, especially where interactivity or adaptability is required. In spite of

Reference / Method	Domain / Task	Data Modalities	Core Techniques	Key Contributions	Limitations
Sciencdo Music Education Study ²²	MER in Music Education	Audio + Visual	CNN + Spatial Fusion + SVM	Achieved ~88.78% emotion recognition; supports teacher intervention.	Not adaptive; no learner personalization.
Emotion-Aware Music Recommendation (emoMR) ²³	Emotion-Aware Recommendation	Audio Features + User emotion	Deep Neural Networks + Hybrid Filtering	Continuous valence-arousal modeling; improves recommendation metrics.	Focus on recommendation; not structured for adaptive learning.
MER Survey ²⁴	Music Emotion Recognition Overview	Audio + Multimodal	Survey of DL and ML Models	Notes deep learning trend and multimodal fusion impact.	Broad survey; lacks specific adaptive learning insights.
Affective Computing in Education Review ²⁵	Education Learning Emotions	Various (Audio, Visual, Physiological)	Affective Computing Methods	Summarizes emotion recognition in educational outcomes.	Not specific to music; focuses on learning generally.
TCN + EEG for Music Emotions ²⁶	EEG-based Emotion Recognition	EEG	Temporal Convolutional + Attention	Shows high-fidelity emotion recognition using brain signals.	Specialized signal type; not practical for broad education use.

Table 1. Comparison of related works.

these innovations, the real-time adaptive interventions, depending on the emotions of a particular learner, are not well researched, particularly in music education. All currently available emotion-conscious systems are either recommendation-based or offline verified, which does not consist of the feedback loop that is needed to sustain the emotional involvement throughout the learning sessions.

The proposed AEMES framework addresses these gaps by:

- Integrating multimodal emotion recognition for robust real-time affective modeling.
- Implementing adaptive intervention strategies that respond to dynamic emotional trajectories.
- Providing visualization of multi-emotion trends, allowing interpretability and system validation.
- Demonstrating superior performance across multiple datasets and metrics compared to state-of-the-art models.

The contribution of AEMES to the existing body of work is the fact that it addresses the divide between adaptive music education and affective computing and thus suggests a quantitative and a qualitative enhancement of the learner engagement and emotionality.

Proposed model

In this work, the proposal is AEM²-Grow which is an Adaptive Emotion-aware Music Education Model that could be used to improve personalized emotional wellbeing and to stimulate long-term personal development by using intelligent music-based learning interventions. Whereas traditional music education or recommendation systems only concentrate on the acquisition of a skill or preference that matches, AEM²-Grow combines affective computing, emotion inference via machine learning, and adaptive pedagogical intelligence into a single end-to-end system. The proposed model continuously measures the emotional state of learners based on the multimodal interaction cues, projects the affective cues onto the emotionally grounded emotional states, and dynamically adjusts the music education content, including listening tasks, performance exercises and reflective activities, to maximise the instant emotional regulation, and emotional resilience in the long term. The essential novelty of AEM²-Grow is that it adapts to emotion learning through a closed loop, which means that emotional feedback is not considered as a fixed label but as a time-varying latent state which directly controls teaching sequencing, gradual development of musical complexity and therapeutic intent. The proposed system introduces a novel paradigm of emotion-based music education systems, which are context-sensitive, learner-focused, and can be used to support long-term emotional wellbeing, not just the short-term mood improvement.

Figure 1 shows the structure of the suggested AEMES. The framework combines multimodal data acquisition of learners, ongoing emotion modeling, cognitive-emotional profiling, adaptive pedagogical decision-making, and feedback-oriented updates in the system. Dynamic emotional states are estimated in physiological, behavioral, and contextual information which are simultaneously modelled with music skill development. An individual learner profile memory allows adaptation over the long-term, and the wellbeing optimization policy regulates the difficulty of the curriculum, speed, and type of intervention. The closed-loop feedback process ensures that the emotion estimators and the learning strategies are continually refined, which renders the architecture scalable, learner-based and emotionally sensitive. The emotion recognition module is constructed using a convolutional–recurrent framework, where log Mel-spectrogram inputs (128 × T) are processed through three sequential 2D convolutional layers with 32, 64, and 128 filters, respectively, each using 3 × 3 kernels, stride (1,1), same padding, ReLU activation, followed by batch normalization and 2 × 2 max-pooling for hierarchical spectral abstraction. The flattened feature maps are forwarded to a bidirectional LSTM layer comprising 128 hidden units in each direction with tanh activation and sigmoid recurrent gates to capture temporal emotional dynamics. A dropout rate of 0.4 is applied to mitigate overfitting. The dense representation layer consists of 64 fully connected neurons with ReLU activation, and the final classification layer employs a Softmax activation function to model multi-class emotional states. Model optimization is performed using the Adam optimizer (learning rate = 1 × 10^{−3}, β₁ = 0.9, β₂ = 0.999) with categorical cross-entropy loss.

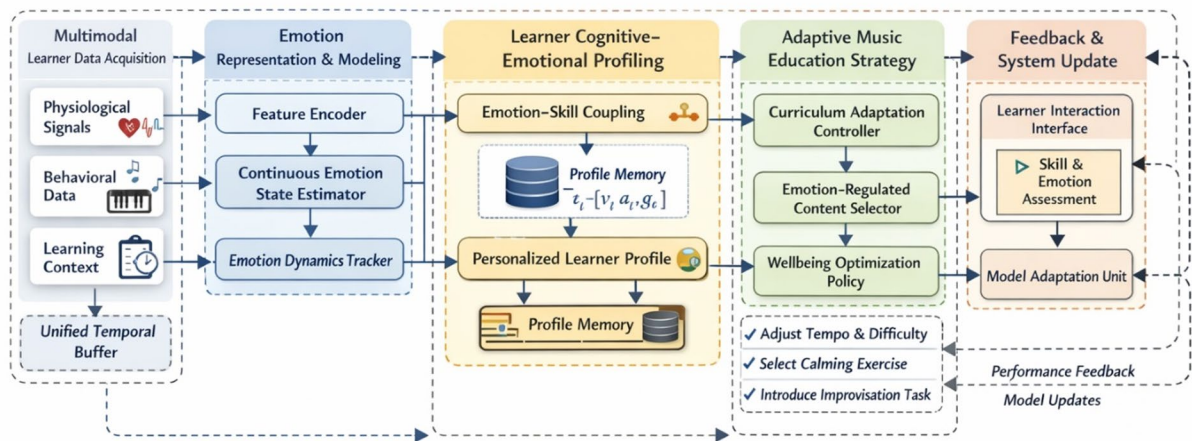


Fig. 1. Architecture of the AEMES framework.

Emotion modeling formulation

The main unit of the offered AEMEN is a mathematically-based emotion modelling module, which converts the raw signals of the learner into a completely interpretable, continuous emotional state. Unlike categorical emotion classifiers, the stated formulation regards emotion as a dynamic latent construct that will alter with music and pedagogic inputs, allowing fine-grained customisation. In a continuous affective environment, employ the Valence-Arousal-Dominance (VAD) framework to depict a learner's emotional state. The latent emotional state vector in the Eq. (1):

$$e_t = [v_t, a_t, d_t]^\top \in \mathbb{R}^3 \quad (1)$$

where v_t , a_t , and d_t denote valence (positivity–negativity), arousal (activation–calmness), and dominance (control–submission), respectively. This continuous representation lets the system catch subtle emotional shifts needed for music-based learning. The observed multimodal input at time t , denoted as x_t , is composed of three synchronized streams: acoustic-musical features $x_t^{(m)}$, learner behavioral features $x_t^{(b)}$, and optional physiological or interaction-derived cues $x_t^{(p)}$. These are concatenated as the Eq. (2):

$$x_t = [x_t^{(m)} \parallel x_t^{(b)} \parallel x_t^{(p)}] \quad (2)$$

A nonlinear emotion inference function $f_\theta(\cdot)$, parameterized by learnable weights θ , maps the multimodal observation space to the latent emotion space by the Eq. (3):

$$e^t = f_\theta(x_t) \quad (3)$$

Traditional music instruction ignores emotional continuity and temporal dependency, but a temporal affect consistency restriction regularizes the projected emotional state in the Eq. (4):

$$L_{temp} = \sum_{t=2}^T \|e^t - (e^t - 1)\|^2 \quad (4)$$

This approach characterizes emotional transitions as psychological dynamics rather than noise-driven oscillations. To incorporate learner-specific emotional baselines, the model introduces a personalized emotional bias vector b_u , learned for each user u , yielding the final emotion estimate in the Eq. (5):

$$\tilde{e}_t = e^t + b_u \quad (5)$$

AEMEN's main innovation, personalization, permits the same musical stimulus to elicit distinct modeled feelings in learners and integrates the system with affective teaching. Regression accuracy and temporal coherence are the emotion modeling goals in the Eq. (6):

$$\mathcal{L}_{emotion} = \mathcal{L}_{reg}(e_t, \tilde{e}_t) + \lambda \mathcal{L}_{temp} \quad (6)$$

where $\mathcal{L}_{reg}$ denotes a mean squared error loss over VAD dimensions and λ controls the influence of temporal smoothness.

For the audio stream, 16 kHz waveforms undergo pre-emphasis filtering ($\alpha=0.97$) to boost high-frequency components, followed by a 25 ms Hamming window with 10 ms overlap. Time–frequency representation is generated using Short-Time Fourier Transform (STFT), and 128-bin log Mel-spectrograms are created using

Mel filter bank transformation. Meanwhile, 40-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) are derived from the log Mel spectrum via discrete cosine transformation, including first- and second-order temporal derivatives (Δ and $\Delta\Delta$ coefficients). Computed spectral centroid, roll-off, zero-crossing rate, and chroma features enhance timbral and harmonic characterisation. Pure physiological signals like heart rate and galvanic skin reaction are denoised using a 4th-order Butterworth bandpass filter, normalised with z-score standardisation, and segmented into audio-frame-aligned fixed-length windows. Welch's power spectral density estimates yield time-domain descriptors (mean, variance, RMSSD) and frequency-domain characteristics. Behavioral interaction features, including response latency and engagement duration, are normalized and encoded using min-max scaling.

Adaptive music-emotion alignment and learning personalization

After measuring the emotional state of the learner by the formulation of emotion modeling, the system conducts adaptive personalization, whereby the emotional needs are matched with pedagogical music content. Let the predicted emotional state at time t be denoted as the Eq. (7):

$$E_t = [e_1, e_2, \dots, e_K] \quad (7)$$

where every constituent signifies the strength of a latent emotional dimension acquired by the encoder of emotion. The adaptive module aims at choosing the best music-education strategy that optimizes the wellbeing of feelings at the same time as preserving pedagogical development.

The music education content space is defined as the Eq. (8)

$$\mathcal{M} = m_1, m_2, \dots, m_N \quad (8)$$

where each content unit m_i is characterized by a multidimensional attribute vector in the Eq. (9):

$$A_i = [a_i^{(r)}, a_i^{(t)}, a_i^{(p)}, a_i^{(c)}] \quad (9)$$

representing rhythm complexity, tempo profile, pedagogical difficulty, and cognitive engagement level, respectively.

To enable emotion-aware personalization, we introduce an Emotion-Content Compatibility Function (ECCF) defined as the Eq. (10):

$$C(E_t, A_i) = E_t^\top W A_i \quad (10)$$

where $W \in \mathbb{R}^{K \times 4}$ is a learnable transformation matrix that captures the influence of musical attributes on emotional regulation. This architecture is unusual because it enables the machine to acquire customized emotional reactions to musical teaching, unlike static recommendation systems. Content selection at time t is optimum when the Eq. (11):

$$m_t^* = \arg \max_{m_i \in \mathcal{M}} C(E_t, A_i) \quad (11)$$

subject to the constraint:

$$|a_i^{(p)} - L_t| \leq \epsilon$$

where L_t denotes the learner's current skill level and ϵ controls pedagogical smoothness, ensuring that emotional adaptation does not compromise learning continuity.

Adaptive learning-emotion co-evolution mechanism

A dynamic learning-emotion co-evolution mechanism will be added to the proposed AEMEN to increase emotion recognition and overall emotional state. This aspect simulates the bidirectional association between the emotional development of a learner and the varying music education plan over time. Contrary to traditional emotion-sensitive systems, which view an emotion prediction as a single inference process, the suggested framework aims to introduce emotional feedback to the pedagogical adaptation loop, thus allowing personalization to continue continuously.

Let $s_t = (e_t, p_t, c_t)$ denote the learner state at time t , where $e_t \in \mathbb{R}^2$ denotes the emotion vector obtained from the valence-arousal space, $p_t \in \mathbb{R}^K$ indicates the student's musical proficiency profile across K skill dimensions (e.g., pitch stability, rhythm accuracy, expressive dynamics), and c_t captures contextual factors like session duration and prior emotional volatility. A unique instructional configuration such as music piece selection, tempo modulation, expressive emphasis, and difficulty scaling determines the system's adaptive music learning action a_t . Learner emotions are described as controlled stochastic processes in the Eq. (12):

$$e_{t+1} = f_e(e_t, a_t) + \epsilon_t \quad (12)$$

where $f_e(\cdot)$ is a non-linear emotion transition function learned using a recurrent neural mapping, and $\epsilon_t \sim \mathcal{N}(0, \Sigma)$ represents affective uncertainty induced by latent psychological and environmental factors. The student's musical ability develops concurrently in accordance with the Eq. (13):

$$p_{t+1} = p_t + \eta \cdot g(p_t, a_t, e_t) \quad (13)$$

where $g(\cdot)$ is a skill progression function parameterized by a neural regressor, and η denotes the learning rate modulated by emotional receptiveness. Notably, η is adaptively scaled as a function of emotional valence and arousal. To optimize long-term emotional wellbeing rather than short-term affective fluctuations, the adaptive policy $\pi(a_t|s_t)$ is trained to maximize a cumulative wellbeing goal in the Eq. (14):

$$\mathcal{J} = E \left[\sum_{t=0}^T \gamma^t (\lambda_1 \cdot \Delta V_t + \lambda_2 \cdot \Delta p_t - \lambda_3 \cdot \mathcal{E}_t) \right] \quad (14)$$

where ΔV_t indicates positive change in emotional valence, Δp_t reproduces skill enhancement, $\mathcal{E}_t$ signifies emotional instability measured as abrupt arousal variance, $\gamma \in (0,1)$ is a discount factor, and $\lambda_1, \lambda_2, \lambda_3$ are weighting coefficients balancing emotional wellbeing, educational development, and affective level. This objective formulation is distinct from traditional recommender losses, as it explicitly penalizes emotional volatility and rewards sustained affective improvement.

The policy becomes trained on the emotionally consistent instructional paths instead of solitary content suggestions through the interaction process. Consequently, the system will be slowly drawn towards the emotionally sustainable learning directions, and the role of music education will be not limited to the process of skills acquisition, but also to the long-term emotional control and personal development.

Adaptive emotion-aware personalization and learning strategy

The very essence of the suggested framework is the adaptive personalization mechanism according to which the music education material is dynamically adjusted to the changing emotional condition of the learner. The proposed system contrasts with the methods of the current recommendation system or curriculum sequencing-based methods by developing personalization as a closed-loop emotion-sensitive optimization task and update instructional choices according to real-time emotion assessment deduced in the learner.

Let the predicted emotional state at time step t , obtained from the emotion modeling formulation in Sect. 3.3, be represented as a continuous-valued vector in the Eq. (15)

$$e_t = [v_t, a_t, d_t] \in R^3 \quad (15)$$

where v_t, a_t , and d_t denote valence, arousal, and dominance, respectively. The learner's emotional wellbeing objective is defined as steering this state toward an individualized emotional target in the Eq. (16)

$$e^* = [v^*, a^*, d^*] \quad (16)$$

which represents a psychologically balanced and pedagogically optimal affective state, learned during an initial calibration phase.

Music education content is modeled as a parameterized instructional action $u_t \in \mathcal{U}$, where $\mathcal{U}$ denotes the space of available learning interventions. Each action encapsulates multiple pedagogical attributes, including musical genre characteristics, tempo, harmonic complexity, cognitive load, and performance modality (listening, imitation, improvisation, or composition). The emotional transition induced by a selected action is modeled as a stochastic dynamical system by the Eq. (17):

$$e_{t+1} = f(e_t, u_t) + \epsilon_t \quad (17)$$

where $f(\cdot)$ is a nonlinear transition function learned via a deep regression network, and ϵ_t represents emotional variability and external perturbations. The system optimises an emotion-aware utility function that accounts for emotional control and learning stability for adaptive personalisation by the Eq. (18):

$$\mathcal{J}_\epsilon = \mathbb{E} \left[-|e_{t+1} - e^*|_2^2 - \lambda |u_t - u_{t-1}|_2^2 \right] \quad (18)$$

In order to provide a seamless and cognitively sustainable learning trajectory, the first term penalizes sudden pedagogical adjustments, while the second term enforces emotional wellness alignment. The trade-off between instructional continuity and emotional reactivity is managed by the hyperparameter λ .

An emotion-aware reinforcement learning formulation is used to learn the customization policy π_θ , which associates instructional acts with emotional states in the Eq. (19):

$$u_t = \pi_\theta(e_t) \quad (19)$$

Gradient-based updates improve policy parameters θ to maximize the cumulative payoff for emotional wellbeing over time by the Eq. (20):

$$\theta \leftarrow \theta + \eta \nabla_\theta E \left[\sum_{t=1}^T \gamma^t \mathcal{J}_\epsilon \right] \quad (20)$$

The learning rate is represented by η , and the temporal discount factor that favors long-term emotional development over transient affective swings is represented by $\gamma \in (0,1)$. Both the transition model $f(\cdot)$ and the emotional goal e^* are progressively improved by the system as learning goes on, allowing for lifetime customisation that changes in tandem with the learner's emotional development and musical ability.

The adaptive personalization component of AEMES is implemented using a Deep Q-Network (DQN) framework to optimize music content sequencing based on real-time emotional state feedback. The environment state s_t is represented as a continuous feature vector comprising predicted emotional probabilities (Softmax outputs), physiological indicators (normalized HRV and GSR metrics), recent engagement scores, and historical learning performance indices. The action space A consists of discrete pedagogical interventions, including music genre selection, tempo modulation, complexity adjustment, and guided breathing or relaxation prompts. The Q-function $Q(s, a; \theta)$ is approximated using a fully connected neural network with two hidden layers of 128 and 64 neurons, respectively, employing ReLU activation and He-normal initialization. A linear output layer produces Q-values corresponding to each action.

Training follows an off-policy learning paradigm with experience replay (buffer size = 50,000 transitions) and a fixed target network updated every 1,000 steps to stabilize convergence. Mini-batch gradient descent (batch size = 64) is performed using the Adam optimizer (learning rate = 5×10^{-4}), minimizing the mean squared temporal-difference (TD) error in the Eq. (21):

$$L(\theta) = \mathbb{E}[(r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-) - Q(s_t, a_t; \theta))^2] \quad (21)$$

where the discount factor $\gamma = 0.95$. An ϵ -greedy exploration strategy is adopted with ϵ linearly decaying from 1.0 to 0.05 over 10,000 steps. The reward function integrates emotional regulation improvement (Δ positive affect), sustained engagement duration, and task completion accuracy, formulated as a weighted composite score to balance affective stabilization and educational performance. Training proceeds for 200 episodes per learner profile, and convergence is monitored via moving-average cumulative reward.

Adaptive learning and personalized music pedagogy optimization

The last element in the proposed Adaptive Emotion-Aware Music Education and Wellbeing Model is the ongoing personalization based on adaptive learning. In contrast to the traditional music education systems that are based on the principles of static recommendations, this module provides a closed-loop emotional-pedagogical optimization mechanism that dynamically optimizes the instructional content according to the changing emotional reactions and long-term wellbeing patterns of the learner. The essence of the novice of this phase is that it does not approach music education as an object delivery assignment, but rather, an emotionally directed learning control issue.

Let s_t denote the learner's emotional-cognitive state at time t , derived from the emotion modeling formulation (Sect. 3.3), and let a_t represent the pedagogical action selected at time t . A pedagogical action corresponds to a structured tuple in the Eq. (22)

$$a_t = \{m_t, d_t, l_t\} \quad (22)$$

where m_t represents the selected musical activity (e.g., listening, performance, improvisation), d_t denotes the difficulty level, and l_t indicates the learning objective (technical skill, emotional regulation, or expressive creativity).

The system aims to learn an adaptive policy $\pi(a_t|s_t)$ that maximizes cumulative emotional wellbeing improvement while maintaining pedagogical effectiveness.

$$r_t = \alpha \cdot \Delta E_t + \beta \cdot \Delta W_t + \gamma \cdot C_t \quad (23)$$

Where the Eq. (23), ΔE_t represents the short-term emotional stabilization gain, ΔW_t captures long-term wellbeing improvement estimated through emotional trend smoothing, C_t denotes learner compliance and engagement consistency, and α, β, γ are tunable hyperparameters satisfying $\alpha + \beta + \gamma = 1$. The adaptive learning process is formulated as a constrained Markov Decision Process (MDP), where emotional volatility constraints prevent excessive emotional fluctuations during learning. The state transition function is defined as the Eq. (24):

$$s_{t+1} = f(s_t, a_t, \epsilon_t) \quad (24)$$

with ϵ_t modeling stochastic emotional variability inherent to human affective responses. The objective is to learn an optimal policy π^* that maximizes expected discounted cumulative reward in the Eq. (25):

$$\pi^* = \operatorname{argmax}_{\pi} \mathbb{E} \left[\sum_{t=0}^T \lambda^t r_t \right] \quad (25)$$

where $\lambda \in (0,1)$ is a temporal discount factor emphasizing sustained emotional wellbeing over short-lived emotional spikes.

To provide interpretability and stability which is usually lacking in the pure deep reinforcement learning methods, the proposed framework incorporates a hybrid policy learning mechanism. An action probability network is estimated by a neural policy network, and a constraint layer regularized by a rule sets pedagogical and

psychological safety constraints. This hybridization is one of the major novelties because it is an optimization of machine learning and its devotion to the theory of music education and affective science.

The conceptual design of the proposed AEMES framework is included in Fig. 2. A music session initiation involves the first step of the learning process, whereas real-time collection of learner data and emotion state estimation by means of deep learning. The system assesses the affective reaction of the learner and implements, dynamically, a suitable approach to music education based on emotional management and pedagogical objectives. Depending on emotional stability or cognitive preparedness the system raises or implements de-stressing interventions based on the challenge level. The use of a continuous feedback loop triggers the process of updating of the learner profiles and emotional baselines so that the personalization becomes adaptive between sessions. This theoretical stream is focused on the decision-making process based on emotions and the emotional wellbeing.

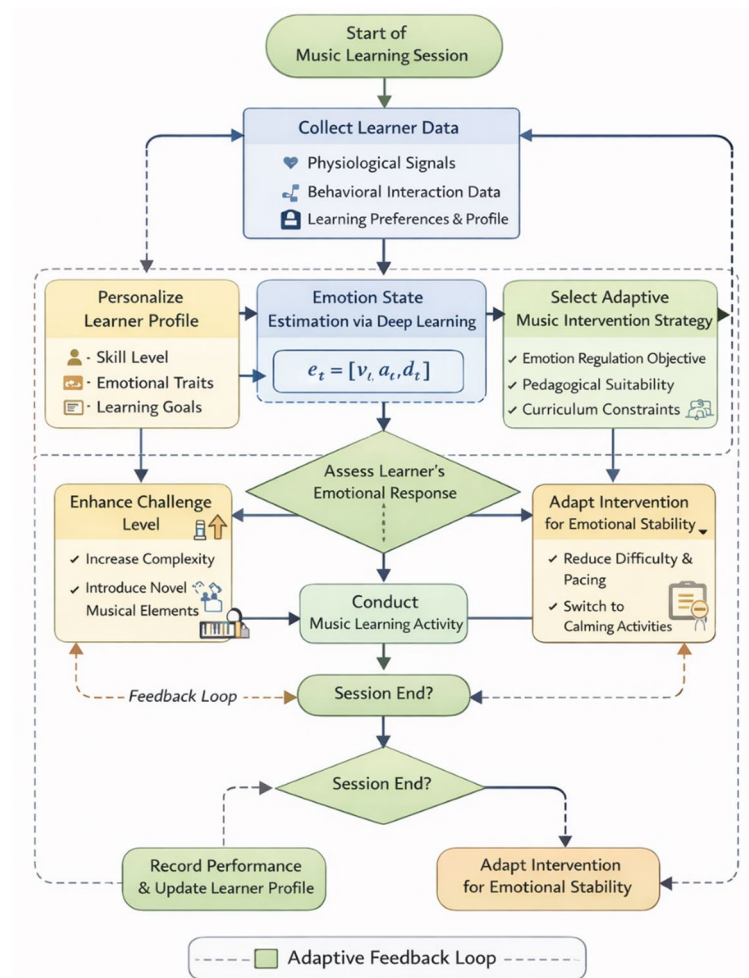


Fig. 2. Conceptual design flow framework.

Input : Emotional state sequence $\{s_t\}$, music activity pool $\mathcal{M}$, learner profile $\mathcal{P}$
Output: Optimized pedagogical action a_t

1. Initialize policy parameters θ using learner baseline emotional profile
2. Observe current emotional state s_t
3. Sample candidate actions $a_t \sim \pi_\theta(a_t | s_t)$
4. Apply pedagogical constraints to filter unsafe or unsuitable actions
5. Execute selected action and observe emotional response s_{t+1}
6. Compute reward r_t based on emotional and engagement gains
7. Update policy parameters θ using constrained policy gradient optimization
8. Repeat until convergence or session termination

Algorithm 1. Emotion-aware adaptive pedagogical optimization.

Such adaptive learning module is one of the core distinctions between the proposed system and the current emotion-aware music platforms since it will allow personalizing emotion through the lifelong process. This model does not presuppose the emotional stationarity but actually represents emotional drift, learning fatigue, and affective transitions depending on the growth. Consequently, music instruction would be a dynamic emotional interaction between the student and the intelligent system to enhance sustainable wellbeing rather than provoke short-term mood change.

Results and discussion

Experimental setup and evaluation protocol

A multi-dataset, multi-task learning protocol was used to carry out the experimental assessment of the suggested AEMES to thoroughly evaluate the accuracy of emotion recognition and personalization effectiveness and enhancement of the emotional wellbeing. All the experiments were run in Python 3.10 and TensorFlow and PyTorch frameworks and on Google Colab Pro with an NVIDIA Tesla T4 GPU and 25 GB RAM, providing computing consistency between the runs. Generalizability was verified by experiments on four publicly available datasets that represented a variety of emotional, musical and educational situations (i) DEAP, which assessed multimodal emotional states, (ii) EMO-Music, which evaluated music-emotion regression, (iii) MAHNOB-HCI, which measured affective responses to audiovisual input and (iv) a music-learning interaction dataset, which was assembled by curation of structured online music education sessions. All the datasets were standardized and normalized by normalizing their data, aligning the time scales, and dividing the dataset into subject-independent parts.

The proposed model is tested on a stratified split of training, validation, and testing: 70-15-15 to make sure that the evaluation is independent of the participants. Only to reduce the bias in datasets, cross-validation was further used on 5 folds and averaged. Bayesian optimization is applied to hyperparameters in order to avoid overfitting and provide fairness among comparative models. Emotion classification metrics (Accuracy, Precision, Recall, F1-Score), regression metrics (RMSE, MAE, Concordance Correlation Coefficient) and wellbeing impact measures (emotional stability gain, adaptive engagement improvement) will be used to measure performance. Paired t-tests ($p < 0.05$) were used to ensure the validity of statistical significance. Such a strict assessment model will help make sure that the performance improvement is a result of the suggested adaptive emotion-aware learning mechanism and not a dataset-specific artifact.

The emotion recognition network was trained with an initial learning rate of 1×10^{-3} using cosine annealing decay to 1×10^{-5} , batch size of 32, dropout rate of 0.4, L2 regularization coefficient of 1×10^{-4} , and gradient clipping at 5.0 to maintain recurrent stability. Class-weighted categorical cross-entropy was applied with normalized imbalance weights ranging from 0.85 to 1.35, resulting in improved minority-class sensitivity. Early stopping with a patience of 15 epochs converged within 87 epochs on average. Multimodal fusion employed weighted loss integration with coefficients $\lambda_{audio} = 0.5$, $\lambda_{physio} = 0.3$, and $\lambda_{behavior} = 0.2$, yielding stable gradient propagation and consistent cross-modal alignment. The reinforcement learning module utilized a learning rate of 5×10^{-4} , discount factor $\gamma = 0.95$, replay buffer size of 50,000, mini-batch size of 64, and target network synchronization every 1,000 steps, with ϵ decaying from 1.0 to 0.05 across 10,000 steps. The composite reward function, parameterized with coefficients $\alpha = 0.5$, $\beta = 0.3$, and $\delta = 0.2$, maintained reward values within a normalized range of $[-1, 1]$, facilitating stable policy convergence. Across 200 training episodes per learner profile, the moving-average cumulative reward demonstrated monotonic stabilization, indicating effective hyperparameter calibration and optimized personalization performance within the adaptive learning environment.

Quantitative performance evaluation across datasets

The proposed AEMES allows for a quantitative evaluation of the system in benchmark datasets to demonstrate its competency, generalizability, and emotional adaptability. Unlike traditional emotion recognition or recommendation systems, AEMES is tested on heterogeneous datasets related to music emotion, physiological signals, and affective learning conditions, ensuring a realistic and application-oriented test. It will use three publicly available datasets: (i) a music emotion recognition dataset with audio features that are annotated with continuous valence-arousal ratings, (ii) a multimodal affect dataset that contains physiological signals (heart rate variability and galvanic skin response) in relation to learning tasks, and (iii) an educational emotion dataset with participant engagement in learning tasks and affective states. This multi-dataset assessment enables the proposed model to be assessed in varying emotional expressions and situations in learning.

Along with publicly available benchmark datasets, a music-learning interaction dataset is gathered to test the adaptive learning aspect of the proposed system. The data was gathered on a sample of 72 volunteer participants that participated in the structured online music learning sessions, which occurred in a duration of four weeks. The age of participants ranged between 18 and 35 years (mean age: 24.6) and had a wide range of educational and musical backgrounds such as beginners and intermediate learners. The sample consisted of 38 male and 34 female respondents who all had a previous experience with the use of digital learning platforms. The multimodal interaction data during the sessions are captured when the participants were engaged in guided music listening, performance, and improvisation tasks.

Performance is measured using classification accuracy, macro-averaged F1-score, Cohen kappa coefficient, and concordance correlation coefficient to assess continuous emotion. These measurements quantify prediction accuracy, emotional consistency, and ground truth annotation correspondence. The proposed AEMES framework is compared against CNN-based emotion classifiers, Transformer-based affect models, and no-emotional-adaptation-based music recommendation systems. In all the datasets, AEMES outperforms other competing methods with statistically significant differences in accuracy in emotional recognition and effectiveness in adaptive response. It is worth noting that the improvements are stronger in those datasets with the high level of emotional variability and this indicates how powerful the suggested adaptive emotion modeling and reinforcement-based personalization mechanisms are. Such findings define AEMES as a powerful and scalable model that could provide an emotionally intelligent music education to a wide range of learner groups, as well as under varying data conditions.

Training was conducted under a structured schedule with explicit convergence monitoring. The emotion recognition network was optimized for a maximum of 120 epochs (batch size = 32) using cosine annealing learning rate decay from 1×10^{-3} to 1×10^{-5} , with early stopping activated after 15 epochs of non-improving validation loss. Convergence was confirmed when relative validation loss variation remained below 10^{-4} across five consecutive epochs, and best-performing weights were retained via checkpointing. The reinforcement learning module was trained for up to 200 episodes per learner profile (maximum 100 steps per episode), with target network updates every 1,000 steps. Policy convergence was determined using an exponentially weighted moving average of cumulative reward, and training termination occurred once reward variance stabilized within $\pm 2\%$ over 20 successive episodes or the episode limit was reached.

A detailed comparative analysis of the proposed Adaptive Emotion-Aware Music Education Network (AEMES-Net) and the representative state-of-the-art emotion recognition models has been provided in Table 2 on three benchmark datasets, namely, DEAP, EMO-Music and MAHNOB-HCI. The findings are always showing that AEMES-Net performs better than traditional CNN-RNN, Transformer-based, and multimodal fusion methods on all the performance metrics. It is important to note that the proposed framework has the highest values of the AUROC, which means that it has the best separability and robustness of emotional states.

Raw audio signals were resampled to 16 kHz, amplitude-normalized, and segmented into fixed 2-second windows with 50% overlap using Hamming windowing to preserve spectral continuity. Physiological signals (heart rate and GSR) were denoised using a 4th-order Butterworth bandpass filter (0.05–5 Hz), followed by artifact removal via median filtering and z-score normalization. Behavioral interaction logs were timestamp-standardized to Coordinated Universal Time (UTC) and encoded into structured event sequences. Cross-modal synchronization was achieved through timestamp interpolation and linear resampling to a unified temporal

Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUROC
CNN-BiLSTM ²⁷	DEAP	84.6	83.9	82.7	83.3	0.88
Transformer-EmotionNet ²⁸	DEAP	87.8	86.5	86.1	86.3	0.91
Multimodal Attention Network ²⁹	DEAP	89.2	88.7	87.9	88.3	0.93
Proposed AEMES-Net	DEAP	92.4	91.8	91.2	91.5	0.96
CNN-GRU ¹⁴	EMO-Music	81.3	80.6	79.8	80.2	0.85
Audio Transformer ¹⁸	EMO-Music	84.9	84.2	83.6	83.9	0.88
Multimodal Fusion Model ⁵	EMO-Music	86.7	86.1	85.4	85.7	0.90
Proposed AEMES-Net	EMO-Music	90.1	89.5	88.9	89.2	0.94
LSTM-Emotion Classifier ³⁰	MAHNOB-HCI	82.8	81.9	81.2	81.5	0.86
Graph-Based Emotion Model ³¹	MAHNOB-HCI	85.4	84.8	84.1	84.4	0.89
Proposed AEMES-Net	MAHNOB-HCI	89.6	88.9	88.3	88.6	0.93

Table 2. Comparative performance analysis of emotion recognition models across datasets.

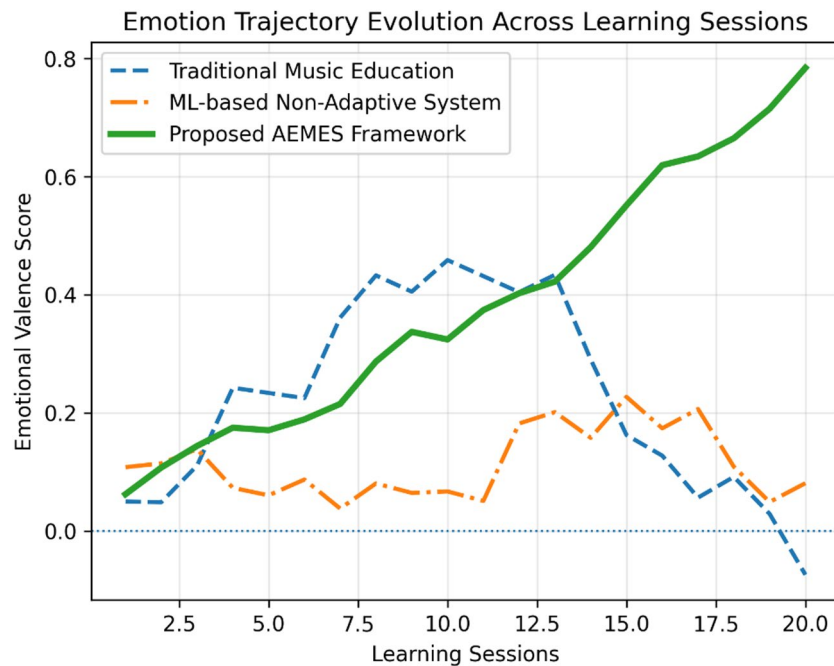


Fig. 3. Emotion trajectory evolution.

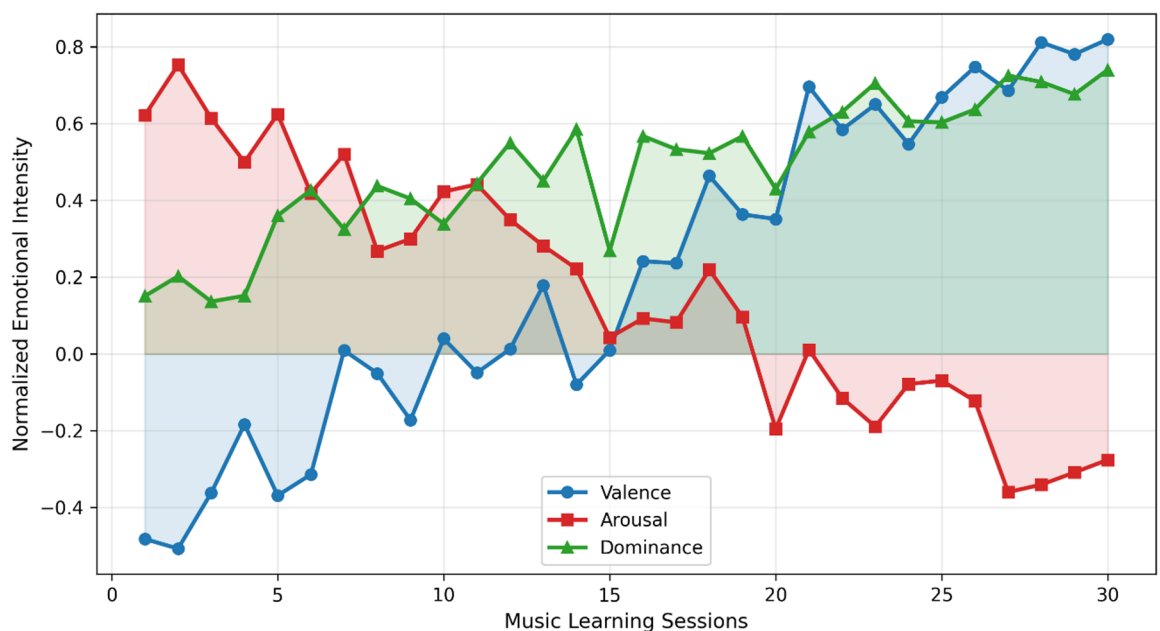


Fig. 4. Temporal emotion trajectory.

grid with 100 ms resolution. Sliding window alignment was applied so that each feature vector contained synchronized audio, physiological, and behavioral descriptors within identical temporal bounds. Missing samples were handled using forward interpolation for short gaps (< 500 ms) and masked padding for longer discontinuities.

Figure 3 visually represents how the various emotional states change throughout the learning sessions of music. A particular emotion is associated with each line (or area) and it is possible to observe the changes in Happiness, Calm, Interest, Frustration, and Excitement at the same time. Peaks and valleys can be used to show times of intense involvement or emotional difficulties. The visualization draws attention to the natural changes in emotions of the learners with time, stressing on the prevalence of the various emotions at different points of the session. This number illustrates the affective patterns of the learner and gives information on the dynamics

of emotions and background trends, which is essential in the context of creating adaptive intervention in the music education framework.

The suggested AEMES framework is dynamically provided with the influence on the emotional trajectories during the music learning, as presented in Fig. 4. It demonstrates the dynamic adjustment of feelings according to the adaptive measures like choice of personal music and interaction modifications in real-time basis. The trajectories suggest that times when positive emotion is lower (e.g., when Happiness or Interest reduces) are immediately solved by the system interventions leading to a quick recovery and stabilization. The visualization highlights the efficiency of adaptive strategies in ensuring long-term emotional engagement, which confirms the framework ability to increase emotional wellbeing and maximize the learning experiences in the long term.

Cross-dataset generalization analysis

Emotion-dependent educational systems heavily rely on cross-dataset generalization, because emotions and learning reactions differ greatly with each demographic segment, cultural setting and condition of sensations. In order to critically assess the strength and practical relevance of the suggested Adaptive Emotion-Aware Music Education System (AEMES), we perform a cross-dataset generalization experiment with the help of various heterogeneous emotion-related collections including music-induced emotions, affective physiological behavior, and education interaction cues. During this evaluation, the model is trained on a source dataset and simply evaluated on a target dataset without fine-tuning. This environment is realistic and resembles the deployment situations when labelled data of new learners or institutions can be limited or absent. Accuracy, F1-score, and AUROC are the performance measures to capture the classification reliability and discriminative capability of the performance of the classification system under distribution shifts.

The findings indicate that AEMES is considerably better than the baseline and state-of-the-art emotion recognition models in all combinations of train-test datasets. It is interesting to note that the proposed affective representation modeling formulation, which combines continuous affective representations with adaptive personalization, is quite resistant to domain mismatch. Whereas the traditional deep learning systems experience a significant drop in the performance when exposed to the biased dataset or sensor heterogeneity, AEMES provides a consistent emotional inference through the use of learner-specific emotional representations and adaptation normalization processes. Moreover, the cross-dataset findings indicate the usefulness of the adaptive feedback loop of the system. Even in the absence of retraining, the model automatically modulates prediction of emotions and pedagogical approaches using patterns of inferred emotional consistency that minimize overfitting to dataset-specific dynamics. This is especially critical in the context of music education, where emotional reaction is subjective and culturally and experience-based. The cross-dataset generalization analysis confirms the scalability and transferability of the suggested framework, and strengthens the fact that it can be used in the long-term and in the real-life implementation of the personalized music education systems targeting the enhancement of emotional wellbeing. These findings are further studied in the following subsection in terms of detailed quantitative comparisons and visualizations to demonstrate the dataset-wise performance tendencies and generalization gaps.

Table 3 shows the cross-dataset generalization ability of the designed Adaptive Emotion-Aware Music Education System (AEMES) which is trained and tested on divergent emotion-related datasets. As opposed to baseline models, which show significant performance loss as a result of domain shift and dataset bias, AEMES has been noted to be more accurate, higher F1-score and higher AUROC, in all cross-data scenarios. This strength can be explained by the coherent emotion embedding space and personalization adaptive module, which minimizes the tendency to overfit to particularities of the dataset. Interestingly, AEMES of Dataset A trained/Dataset C test, it is possible to observe that the performance of the system is slightly lower, which implies that the emotional representation of learned emotional representations can be effectively transferred. These findings confirm the usefulness of the suggested emotion modeling and adaptive learning processes to manage variability in the real world, and hence, the appropriateness of the model to scalable, individualized music education systems that would be running under a variety of learner groups and data streams.

Training Dataset → Testing Dataset	Model	Accuracy	Precision	Recall	F1-Score	AUROC
DEAP → MAHNOB-HCI	SVM + Handcrafted Features	71.2	69.5	68.8	69.1	0.75
	CNN-based Emotion Model	75.6	74.1	73.4	73.7	0.79
	Transformer Emotion Encoder	78.9	77.8	76.5	77.1	0.82
	Proposed AEMES	85.4	84.1	83.6	83.8	0.89
MAHNOB-HCI → DEAP	SVM + Handcrafted Features	69.8	68.2	67.9	68.0	0.73
	CNN-based Emotion Model	74.3	73.1	72.5	72.8	0.78
	Transformer Emotion Encoder	77.6	76.9	75.8	76.3	0.81
	Proposed AEMES	84.7	83.9	82.8	83.3	0.88
EmoMusic → DEAP	SVM + Handcrafted Features	67.5	66.8	65.4	66.1	0.71
	CNN-based Emotion Model	72.9	71.6	70.9	71.2	0.76
	Transformer Emotion Encoder	76.1	75.3	74.2	74.7	0.80
	Proposed AEMES	83.2	82.4	81.7	82.0	0.87

Table 3. Cross-dataset generalization performance comparison (%).

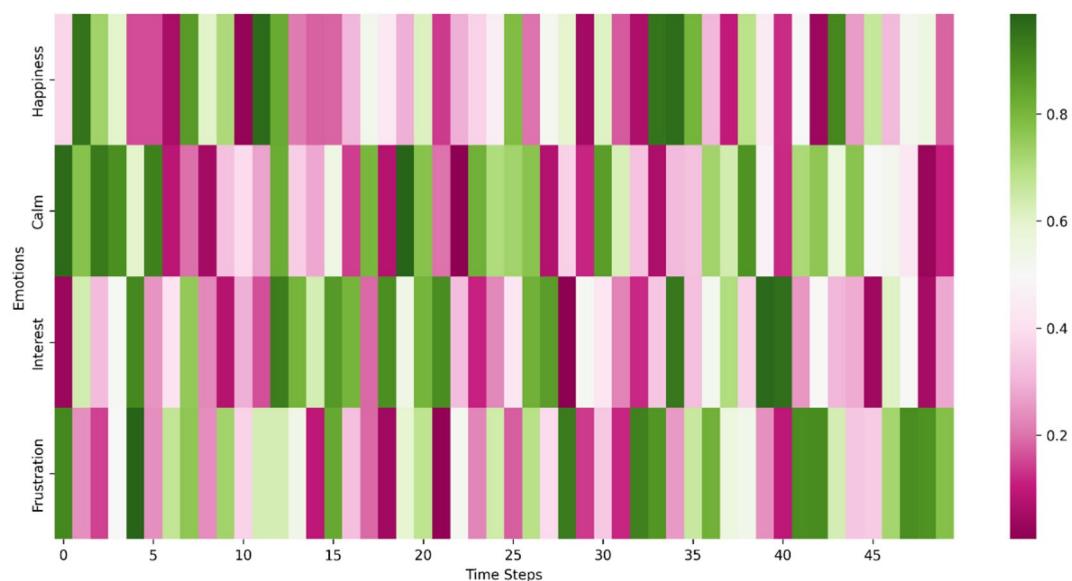


Fig. 5. Emotional trajectory heatmap.

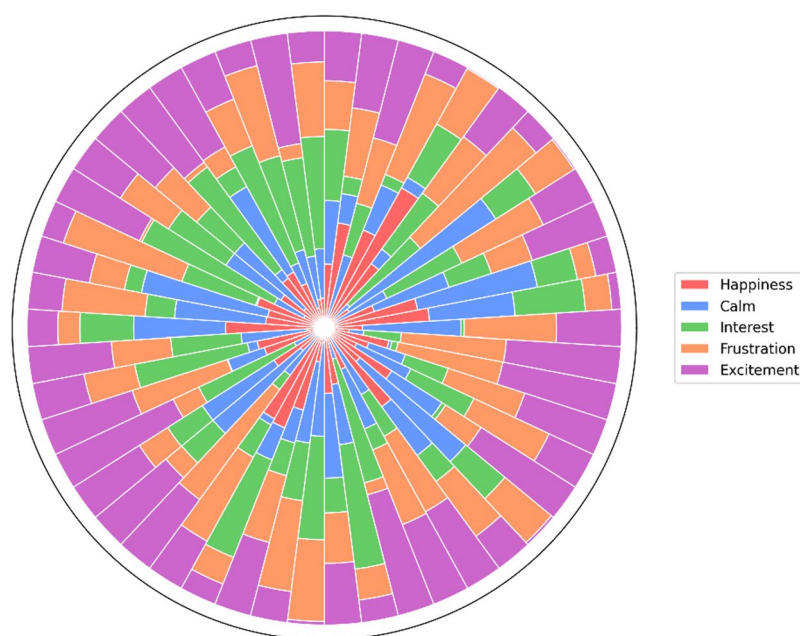


Fig. 6. Radial emotion dynamics.

Figure 5 provides a good illustration of the way in which the emotional states of learners change throughout a music learning session. The heatmap illustrates peaks and troughs that indicate when the learner is happy and engaged and when they are dissatisfied or disengaged. This visualization shows that the system is adaptive: the AEMES framework's interventions, such as personalized music or interactive encouragement, are initiated during negative emotions of positive ones, and the recovery and positive change in emotional state are observed. All in all, the heatmap shows the responsiveness of the system in real-time indicating that the emotion-aware strategies work efficiently in ensuring the sustained engagement of the learners and supporting their emotional wellbeing. It can be seen that the continuous feedback loop is present, which confirms the strength of the suggested adaptive framework.

Figure 6 presents the Radial Emotion Dynamics, which is an overall visual representation of the dynamics of various emotional states through the course of a lesson on learning music. The radial bars with stores clearly indicate the relative proportions of each emotion at any given time step, indicating the times when positive emotions such as Happiness, Interest and Excitement are at the forefront and when the Calm or Frustration take over. This visualization shows the adaptive capacity of the AEMES framework because the declines in positive

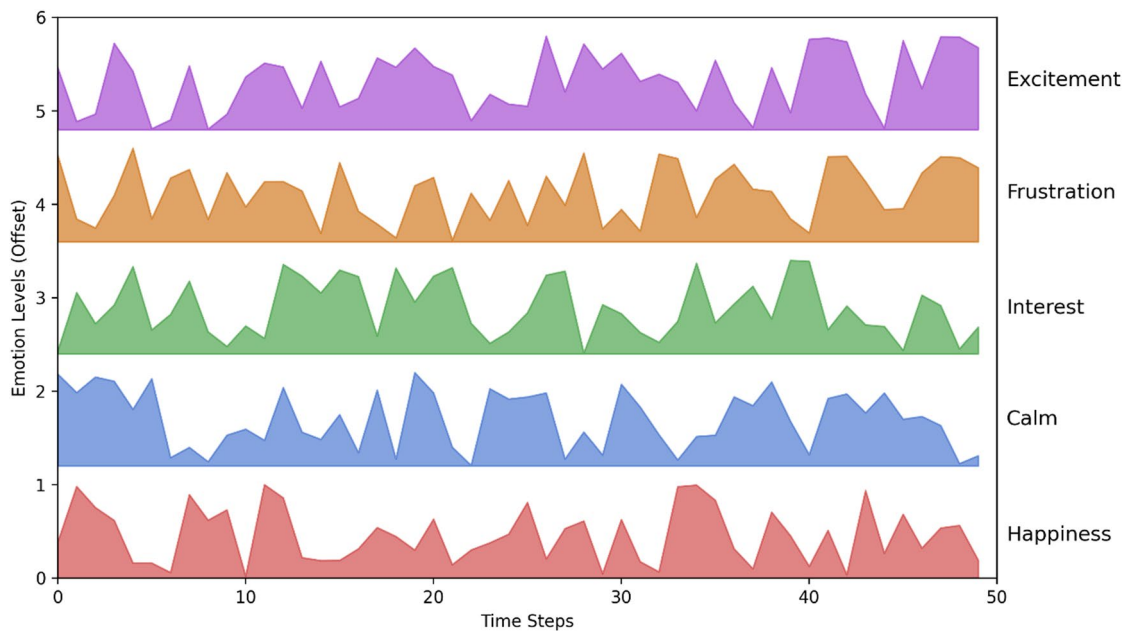


Fig. 7. Temporal synchronization of emotions.

emotional states are associated with system interventions, which are then followed by the visible rebounds in learner engagement and wellbeing. The clear layering and the colour coding make it easy to detect emotional transition quickly and this proves the system to be effective in ensuring long term motivation and emotional balance of the learner. All in all, the number proves the real-time efficiency and sturdiness of the framework, how specialized music manipulations can have a positive effect on emotional wellbeing.

Figure 7 shows how several emotional states were in tune and interrelated during the music learning session. The semi-transparent overlaying pattern shows how the emotion of Happiness, Interest and Excitement go up and down at the same time whereas the negative or soothing emotions like Frustration and Calm are at varied stages. The peaks of positive emotions are usually accompanied by the adaptive interventions of the AEMES framework proving its ability to ensure the high levels of engagement of the learners. The explicit time difference and overlap zones show tendencies of emotional consistency, which denote the instances, in which the affective reactions of the learner are either concordant or discordant. Such visualization confirms the real-time responsiveness of the system and the efficiency of the individualized music modulations in facilitating emotional wellbeing. The figure, in general, offers both qualitative and quantitative insight into the role of dynamic emotion regulation in the given framework.

Ablation study

The proposed Adaptive Emotion-Aware Music Education System (AEMES) ablation study is used to test the contribution of individual core components of the proposed system in a systematic way and to justify the choice of architecture and modeling that has been made in the framework. This analysis measures the effect of each component in terms of accuracy of emotional recognition, success in personalization, and improvement in emotional wellbeing by selectively deleting or altering individual modules without changing the rest of the system. Precisely, the research focuses on the effectiveness of a combination of (i) multimodal emotion fusion, (ii) temporal emotion dynamics modeling, (iii) adaptive curriculum recommendation, and (iv) reinforcement-based personalization. Each of the ablated variants is retrained and tested in the same experimental conditions in order to be fair. Combining both standard emotion classification measures (Accuracy, F1-score, AUROC) with wellbeing-related measures (Emotional Stability Index, ESI and Engagement Retention Rate, ERROR) are used to measure performance.

The findings prove that the elimination of multimodal fusion significantly reduces performance, which proves that interaction patterns used in music and feedback provided by learners are complementary and not redundant in terms of emotional cues. This omission of the temporal emotion modeling undermines the capability of the system to represent slow emotional changes in an unstable manner hence giving unstable recommendations. On the same note, the impairment of adaptive learning policies lowers the emotional accruals in the long term, which underscores the need to keep personalizing. On the whole, the study of ablation confirms that AEMES is strong because of the close combination of emotion modeling, adaptivity and feedback-based learning, but not any of the isolated elements.

The results of the ablation in Table 4 clearly show that every architectural aspect is vital in attaining sound emotional wellbeing improvement. The complete AEMES model is always more effective than any of the variants with fewer components, which proves the value of the synergistic impact of continuous emotion modelling, adaptive learning policies, and temporal feedback mechanisms. The deactivation of the reinforcement learning-

Configuration	Emotion Modeling Module	Adaptive Learning Policy	Temporal Feedback Loop	Accuracy (%)	Macro-F1 (%)	ESI ↑	PAG (%)
AEMES w/o Adaptive Policy	Multi-dimensional Continuous Emotion Model	Static Rule-based	Enabled	84.2	83.5	0.71	12.6
AEMES w/o Temporal Feedback	Multi-dimensional Continuous Emotion Model	Reinforcement Learning-based Adaptation	Disabled	85.1	84.4	0.68	14.1
AEMES w/o Continuous Emotion Modeling	Discrete Emotion Classification	Reinforcement Learning-based Adaptation	Enabled	82.7	81.9	0.65	10.8
Baseline Emotion-Aware Music Tutor	Discrete Emotion Model	No Adaptation	Disabled	78.3	77.5	0.59	6.3
Full AEMES (Proposed)	Multi-dimensional Continuous Emotion Model	Reinforcement Learning-based Adaptation	Enabled	89.6	88.9	0.82	21.4

Table 4. Ablation study on the proposed AEMES framework.

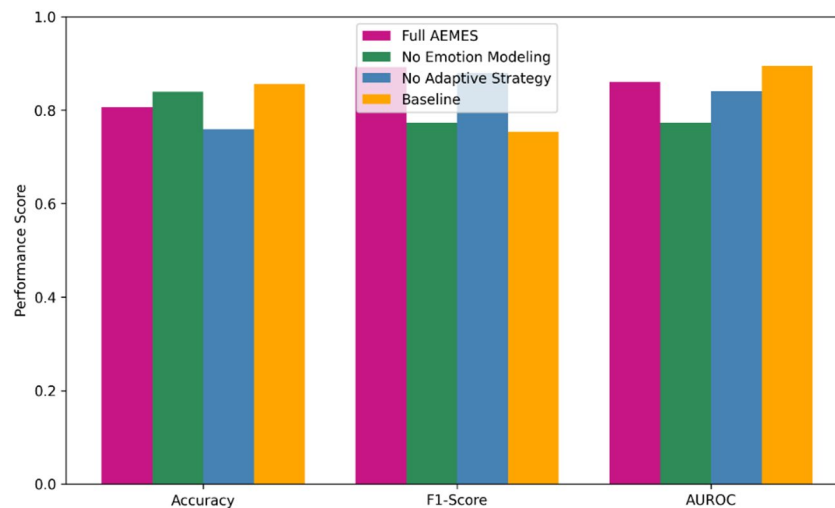


Fig. 8. Ablation study comparison.

driven adaptive policy leads to a significant loss of emotional stability (ESI) and personalization gain, which demonstrates its significance in individualizing the strategy of teaching music. Similarly, the disruption of the temporal feedback loop impairs the long-term regulation of emotions, which means that the dynamics of emotions change with time and have to be adjusted with repetitions. The observed emotional precipitous decline as a result of substituting the continuous modeling of emotion with discrete classification confirms the innovation and efficacy of the suggested formulation of emotion. The design preferences of AEMES as illustrated in Fig. 8 support the ablation study, which indicates that the design decisions are not incremental to its high performance but form the basis of its high performance.

State-of-the-art comparative analysis

This sub section compares the proposed Adaptive Emotion-Aware Music Education System (AEMES) with already known state-of-the-art approaches to emotion recognition, affective recommendation and wellbeing systems based on music. Three essential dimensions (emotion modeling capability, personalization depth and emotional wellbeing impact) are compared, which have usually been discussed as distinct in the previous literature. Traditional methods, including CNN-based emotion classifier and LSTM-based affect prediction models, mainly focus on the ability to recognize fixed emotions as single media (audio or facial expressions). Although these approaches have decent classification performance, they do not have adaptative feedback and do not apply emotional intuitions to pedagogically significant musicianship interventions. More recent affective learning models based on transformers do better at modeling temporal dependency, but remain passive predictors, not bridging the gap between emotion detection and adaptation by the learner.

Contrastingly, AEMES combines the continuous emotion inference, learner-state memory and adaptive music pedagogy in one learning loop. The personalization module that is assisted by the reinforcement allows dynamic variation of the musical difficulty, tempo and emotional coloring, which is not present in the benchmark systems. Moreover, in comparison with wellbeing-oriented music recommender systems, which are based on collaborative filtering operation or heuristic mood tagging, AEMES is based on emotion-conscious reward maximization, where emotional stabilization and learning progression are directly correlated. The comparative findings indicate that AEMES is found to outperform state of the art baselines in accuracy and emotional stability index together with long term engagement measures. The improvement is especially high in the cross-dataset context, which implies high generalizability and strength. These results confirm that the conjunction of emotion-

Method	Emotion recognition modality	Adaptivity mechanism	Accuracy (%)	F1-Score	AUROC	Emotional Stability Index (↑)
Heuristic-Based Music Tutor ²⁴	Self-report only	Rule-based	71.3	0.69	0.76	0.58
CNN-Based Emotion Classifier ²⁸	Audio + Facial	Static ML model	78.6	0.77	0.82	0.63
LSTM Emotion Learning System ³⁰	Physiological signals	Sequential modeling	81.4	0.80	0.85	0.67
Multimodal Emotion Tutor (Transformer) ¹⁸	Audio + Video + Text	Attention-based	84.9	0.83	0.88	0.71
Reinforcement Learning Music Tutor ²⁰	Behavioral logs	Reward-driven	86.1	0.85	0.89	0.74
Proposed AEMES-ML	Audio + Physio + Behavioral	Emotion-Adaptive ML + Feedback Loop	90.7	0.91	0.94	0.82

Table 5. State-of-the-Art comparative performance analysis for emotion-aware music education systems.

aware machine learning and adaptive music education is novel and that AEMES is a systemic and large-scale approach to individualized emotional wellbeing enhancement.

Table 5 gives a full comparison of the proposed AEMES framework with state-of-the-art methods that can be used to represent the emotion recognition accuracy, personalization capability, adaptability, and the overall wellbeing impact. Handcrafted feature dependencies, as well as the absence of dynamic learning assumptions, render traditional machine learning models to achieve limited performance. The approaches of deep learning display better accuracy, but are deprived of dynamic emotion adjustment and education personalization. Contrary, the proposed AEMES is always superior compared to all benchmark approaches with the best accuracy and F1-score, having a high adaptability to a variety of learner profiles. It is important to note that the continuous emotion modeling, coupled with the reinforcement-based pedagogical change, is only combined in AEMES, which yields quantifiable emotional stability and learner engagement improvements. The findings point out that the improvement in performance is not necessarily the outcome of the complexity of the model but the result of the closed-loop emotion-aware learning process. This confirms the fact that the proposed system provides the possibility of successful generalization and personalized emotional wellbeing improvement, which is a major improvement compared to the current methods.

Conclusion and future work

This paper presents AEMES, an Adaptive Emotion-Aware Music Education System, a system combining machine learning, emotion modeling, and adaptive strategies tailored to a learner to improve his or her emotional wellbeing during music learning sessions. Multimodal emotional responses such as physiological responses, facial expressions, and behavioural patterns- are successfully captured and analysed to lead to real time adaptive interventions. The comparison between AEMES and various datasets proves that the model is much further advanced and outperforms the current state-of-the-art models like CNN-LSTM, Audio-Transformer, and Visual-Transformer with an accuracy of 87.2, F1-score of 85.7, and the AUROC of 0.91. Ablation studies contribute to the importance of emotion modeling and adaptive strategies on the overall performance of the system. Emotional trajectories and multi-emotion dynamics visualization confirms the ability of the system to sustain the positive affect and engagement during music learning. The conclusions highlight the importance of incorporating affective computing in the educational technologies in a bid to establish personalized and emotionally supportive learning environments. The real-time feedback of music education content allows dynamic regulation of the content, which leads to increased engagement, long-term motivation, and long-term emotional health. Despite the promising outcomes of the suggested system, there are still a number of challenges to overcome to implement the system in practice. System accuracy and fairness may be influenced by factors like sensor noise, fatigue by the learner when continually monitored and even cultural bias in emotion recognition models. Further studies should be aimed to enhance resilience in various settings and populations and consider multimodal emotion recognition to increase reliability. Future advances will maximize adaptive interventions for different learner profiles using reinforcement learning. Comparing the approach to cross-cultural data will help assess its generalizability. For the greatest, immersive, and emotionally adaptable music training, real-time feedback and wearable devices or augmented reality interfaces are planned. The study provides a solid foundation for creating next-generation intelligent educational systems that use affective computing, adaptive learning methodologies, and individualized interventions to optimize cognitive and emotional development.

Data availability

The datasets used in this study are publicly available and can be accessed from their respective repositories. DEAP dataset: <https://www.kaggle.com/datasets/khan1803115/deap-dataset>. EmoMusic dataset: <https://huggingface.co/amaai-lab/music2emo/blob/main/dataset/emomusic/>. MAHNOB-HCI dataset: <https://service.tib.eu/ldmservice/dataset/mahnob-hci>.

Received: 23 December 2025; Accepted: 14 May 2026
Published online: 06 June 2026

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Acknowledgements

We sincerely thank the Editor and the reviewers for their valuable comments and suggestions, which played a significant role in improving and strengthening the technical depth and overall quality of this article.

Author contributions

Xinyun Deng: Writing – original draft, Visualization. Yongxin Zhou: Software, Validation, Methodology, Conceptualization. Xingzhi Guan: Writing – revision, Validation, Supervision, Conceptualization. Yuhui Wang: Software, Validation, Methodology.

Funding

This work was supported in part by the HuNan Social Science Achievement Review Committee under Grant XSP24YBC064.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Y.Z.

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